Observer–Actor Systems: A Sociologically Inspired Framework for Emergent Coordination in Multi-Agent Systems

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Abstract

1 Introduction

Multi-agent systems research has made substantial progress on individual components of the coordination problem. Value decomposition methods (**?**) address credit assignment. Centralized training with decentralized execution (**?**) addresses the tension between global information and local execution. Differentiable communication (**?**) addresses information sharing between agents. Role heterogeneity methods (**??**) address behavioral specialization. Yet these advances have proceeded largely in isolation: each addresses one facet of coordination while holding others fixed or abstracting them away. No existing framework simultaneously accounts for policy learning, state compression, communication, selective attention allocation under resource constraints, reward mediation under partial observability, and the emergence of influence hierarchies—all within a single coherent mathematical structure. This fragmentation is not merely a matter of presentation. It reflects a deeper conceptual gap: without a unified account of how these components interact, their joint dynamics are difficult to analyze, and the system-level consequences of design choices in one component for behavior in others cannot be formally characterized.

A second limitation runs through most existing approaches: they require some form of centralized coordination, either at training time or in execution. CTDE methods rely on a centralized critic during training that becomes a bottleneck at scale and is unavailable in fully decentralized deployment. Communication architectures typically assume a fixed or globally known topology. Attention mechanisms often require global state to determine which agents to attend to. These dependencies are not incidental—they reflect the absence of a principled account of how coordination can emerge from purely local, bilateral interactions without any agent ever accessing the global state of the system.

The closest existing approaches each address a subset of these requirements but fall short of a unified treatment. CTDE methods such as MADDPG (**?**) and QMIX (**?**) introduce a role asymmetry between a centralized critic and decentralized actors, but the critic exists only at training

time, handles reward information alone, and is discarded at execution—leaving the deployed system with no mechanism for ongoing evaluation, state summarization, or attention allocation. Learned communication architectures such as CommNet (**?**) and TarMAC (**?**) address information sharing between agents, but assume symmetric roles, impose no resource constraints on communication, provide no mechanism for reward mediation, and offer no convergence guarantees for the joint learning dynamics. Role-heterogeneous methods such as ROMA (**?**) and RODE (**?**) allow agents to specialize behaviorally, but roles are discovered through auxiliary objectives rather than derived from a principled account of coordination structure, and neither framework addresses communication, state compression, or attention allocation. Social reward approaches (**?**) introduce influence as a shaping signal but maintain symmetric agent architectures and do not formalize the evaluator–executor distinction that we argue is the structural ingredient missing from all of these methods. The gap, in short, is not that any one component is unaddressed—it is that no framework addresses all of them together, within a single mathematical structure, with convergence guarantees that hold throughout.

The gap becomes visible when one notices that another domain has confronted the same challenges—at far larger scale, without centralized control, and without the ability to iterate in simulation. Human social groups coordinate populations of millions, sustain collective behavior under locally inconsistent information, and solve complex tasks that no individual could accomplish alone. They do so through specific mechanisms—role differentiation, reputational feedback, and the recursive adaptation of behavior to signals from reference groups—that operate entirely at the level of bilateral interactions: no agent ever accesses the global state of the system. These mechanisms are the central object of study in *sociology*, which has spent over a century developing formal, empirically grounded theories of exactly the coordination dynamics that MAS research needs. The opportunity lies in the complementarity between the two fields: sociology provides the structural account of how decentralized coordination works that MAS lacks; MAS provides the formal, algorithmic language that sociology rarely produces. Concretely, sociological theory identifies the asymmetric role distinction between agents who *evaluate* and agents who *act* as the structural feature that makes decentralized coordination tractable—and specifies the feedback mechanisms through which this asymmetry gives rise to the three extensions the Observer–Actor framework adds beyond a direct SRA translation: handling imperfect state information, learning following rates for social feedback, and selecting among observers for state information.

We take a concrete step toward realizing this potential. We draw on the *Socially Responsive Agent (SRA) framework* (**?**), a formal model from computational sociology that operationalizes the micro-level mechanisms of social feedback and behavioral adaptation identified by foundational sociological theory—including symbolic interactionism (**?**), structuration theory (**?**), and field theory (**?**). The SRA framework specifies a feedback loop in which agents adapt their behavior in response to socially mediated evaluations from a reference group, and shows how repeated iteration of this loop gives rise endogenously to stable coordination patterns and influence hierarchies (**???**). Crucially, the framework is stated in procedural terms that admit direct algorithmic translation,

and its bilateral structure means that all coordination is mediated through local interactions—no agent ever requires access to global state.

We introduce the *Observer–Actor framework*, a formal translation of the SRA model into the MAS setting. *Observers* monitor the environment, evaluate the actions of others, and transmit feedback; *Actors* select actions on the basis of that feedback and adapt accordingly. This role asymmetry is not merely a modeling convenience. It is the structural feature that makes a unified treatment of the full coordination problem tractable. By making the evaluator–executor distinction explicit and compositional, the framework addresses policy learning, state compression, communication, selective attention allocation, and reward mediation within a single architecture, with each component admitting its own update rule and convergence analysis. The framework is developed across a series of cases of increasing complexity—from a single Observer–Actor pair under full observability to large populations with partial observability and learnable communication—each adding one component while preserving the structure of earlier cases.

The technical ingredients required to construct the Observer–Actor framework — policy gradients, two-timescale stochastic approximation, projected gradient ascent, differentiable communication — have been available to the MARL community for years. None of them is introduced here. And yet the framework does not exist in the literature. We argue this is not an accident. Building this architecture requires knowing what to build, not just how to build it — and that knowledge is not recoverable from the technical literature alone. The MARL community, working from engineering intuitions and benchmark performance, has produced sophisticated solutions to individual components of the coordination problem while leaving their interactions unaddressed. Sociology, which has spent over a century studying how decentralized coordination actually works in systems far larger and more complex than any multi-agent benchmark, identified the structural primitive that makes a unified treatment tractable: a persistent asymmetric distinction between agents who evaluate and agents who act, operating through bilateral feedback, with influence hierarchies and selective attention emerging endogenously from local interactions. The Observer–Actor framework is what becomes visible once that primitive is in hand. The absence of this framework in the existing literature — despite the availability of all the mathematical tools needed to construct it — is itself evidence that the conceptual insight precedes the architecture, and that the insight is a sociological one.

Contributions. Our contributions are as follows:

- *A unified framework.* We introduce the Observer–Actor framework, a MAS architecture that simultaneously addresses policy learning, state compression, communication, selective attention allocation under resource constraints, reward mediation under partial observability, and the emergence of influence hierarchies—the first framework, to our knowledge, to treat all these components within a single coherent mathematical structure.

- *A formal translation of the SRA model.* We operationalize sociological concepts—feedback

loops, reference groups, and reputational dynamics—as algorithmic components with precise update rules, developing this translation across a series of cases of increasing complexity.

- *Convergence guarantees.* We establish convergence guarantees for every case in the framework, individually and in composition, under conditions that include multiple agents, partial observability, budget-constrained attention allocation, and learnable communication—without requiring any form of centralized training or execution.

- *Illustrative empirical results.* We provide experiments designed to build understanding of how each algorithmic component behaves, demonstrating that the update rules produce the qualitative dynamics the framework predicts.

Advantages of the approach. The Observer–Actor framework offers five properties that distinguish it from existing MARL approaches.

1. *Intuitive.* Once the Observer–Actor distinction is in place, the framework feels surprisingly natural — almost inevitable. The idea that some agents evaluate and others act, that evaluative attention is a limited resource, that actors should learn whom to follow based on the quality of the information they receive, and that influence hierarchies should emerge from these local interactions rather than be engineered top-down: none of this requires justification once the primitive is visible. This quality is not cosmetic. Frameworks that feel forced or arbitrary tend to resist extension and fail to generalize gracefully to new settings; frameworks that feel right tend to do so because they have identified the correct structural primitive. The Observer–Actor framework's intuitive character is, we argue, a consequence of its sociological foundations: the mechanisms it instantiates are ones that human social systems have converged on over millennia, and their naturalness reflects that provenance.

2. *Modularity.* The framework is constructed compositionally: each case isolates a distinct function—reward reporting, state compression, attention allocation, observer selection—and can be deployed independently or combined with others. This means the framework can be adapted to a given setting by activating only the components that setting requires, and that each component can be analyzed and understood in isolation before its interactions with others are considered.

3. *Comprehensiveness.* By treating all components of the coordination problem within a single architecture, the framework makes their interactions visible and formally characterizable. Design choices in one component—say, the Observer's compression function—have traceable consequences for the dynamics of others—say, the Actor's policy convergence—in a way that is impossible to analyze when components are developed in isolation.

4. *Fully decentralized.* The framework requires no centralized component at any stage—neither during training nor execution. All coordination emerges exclusively from bilateral, dyadic

interactions: every feedback signal, every attention update, and every compression refinement flows through local Observer–Actor pairs. No agent ever accesses the global state of the system, aggregates rewards across agents, or relies on a shared critic. This distinguishes the Observer–Actor framework categorically from CTDE methods such as MADDPG and QMIX, which require a centralized critic during training that observes global state and aggregates rewards—a component that is unavailable in fully decentralized deployment and becomes a bottleneck at scale. In the Observer–Actor framework, the structures that CTDE engineers top-down—role differentiation, influence hierarchies, selective attention—emerge endogenously from local interactions, without any agent ever requiring information it could not obtain through bilateral exchange.

5. *Theoretical grounding.* The framework provides formal convergence guarantees across all its cases, individually and in composition. To our knowledge, no existing framework provides such guarantees simultaneously for policy learning, learned communication, budget-constrained attention allocation, and reputation dynamics. This mathematical foundation is not an add-on: it is a direct consequence of the two-timescale update structure that the SRA model's feedback loop naturally induces. The convergence proofs build on established stochastic approximation machinery, and we view this as a strength rather than a limitation. The contribution is not that the proofs are hard — it is that the framework is principled enough that clean guarantees are possible at all. Why this is so is explained by the next point.

The modularity, comprehensiveness, decentralized structure, and the fact that convergence proofs follow from standard machinery rather than requiring bespoke mathematics all trace back to a single source: the Observer–Actor role distinction and bilateral feedback structure are the right architectural primitives, and sociology is what identified them as such.

Scope and methodology. The goal of this paper is to introduce the Observer–Actor framework as a new way of thinking about MAS coordination—not to optimize performance on existing benchmarks or compete with specialized methods on specific tasks. Two methodological choices follow from this goal.

First, throughout the framework we adopt i.i.d. state sampling rather than a full Markov decision process formulation. This is a deliberate choice to reduce notational complexity and keep the focus on the conceptual structure of the framework. The i.i.d. assumption isolates the coordination mechanisms of interest—feedback-driven policy adaptation, state compression, and attention allocation—from the additional complexity of temporal credit assignment. Extending the framework to the MDP setting is a natural direction for future work, but doing so here would obscure the core ideas without adding conceptual insight. We revisit this extension in Section **??**.

Second, the empirical results in this paper are designed to *illustrate* the framework's algorithms and provide intuition for how they behave, not to establish state-of-the-art performance. Each experiment is constructed to make one aspect of the framework's dynamics visible: that Observers

concentrate attention on high-quality Actors, that compression functions learn to highlight task-relevant state features, that Actor selection logits converge to favor informative Observers. These are not performance comparisons; they are proofs of concept that the update rules do what the theory says they should. Benchmark evaluation against strong MARL baselines is left to future work, once the framework has been extended to the MDP setting.

2 Related Work

The Observer–Actor framework sits at the intersection of several active lines of research in multi-agent reinforcement learning (MARL): centralized training with decentralized execution (CTDE), differentiable communication, role-specialized coordination, graph- and attention-based interaction architectures, social influence learning, hierarchical reinforcement learning, and mentor–student formulations of policy adaptation. Each of these lines has made substantial progress on specific aspects of the coordination problem. However, they have largely been developed independently, with each framework targeting a specific coordination function while abstracting away others, and most rely on some form of centralized information aggregation—either at training time or in execution. The Observer–Actor framework differs from all of these approaches in a single, precise way: it derives communication, selective attention, reward mediation, role differentiation, and influence dynamics from a single architectural primitive—persistent evaluator–executor asymmetry mediated through bilateral, dyadic feedback—without requiring any centralized component at any stage, training or execution.

Centralized Training with Decentralized Execution. The dominant paradigm for multi-agent coordination is centralized training with decentralized execution (CTDE) **??**. Approaches such as QMIX **?**, MADDPG **?**, MAPPO **?**, QPLEX **?**, and HAPPO **?** introduce an asymmetry between training and deployment: agents execute decentralized policies, but a centralized critic observes global state and aggregates rewards during training. These methods have proven highly effective for stabilizing learning under non-stationarity and partial observability, and represent the current state of the art on many standard benchmarks. However, the centralized component exists only during training and is discarded during execution, leaving the deployed system with no mechanism for ongoing evaluation, selective attention allocation, communication management, or influence formation. More fundamentally, the centralized critic is unavailable in settings where global state cannot be aggregated at training time—a restriction that applies to many real-world deployments. The Observer–Actor framework treats evaluative asymmetry as a persistent structural feature of the deployed system itself rather than as a temporary training scaffold, and requires no centralized component at any stage. Every feedback signal, every attention update, and every policy refinement flows through local, bilateral Observer–Actor interactions: no agent ever accesses information it could not obtain through dyadic exchange.

Differentiable Communication Architectures. A substantial line of work studies emergent communication and differentiable information exchange between agents. CommNet **?** introduced continuous communication via mean-field aggregation; DIAL **?** backpropagates through discrete communication channels; TarMAC **?** introduces targeted communication via learned attention mechanisms; and RIAL **?** learns communication protocols end-to-end using reinforcement signals. More recent work, including ATOC **?** and I2C **?**, introduces communication gating mechanisms that selectively activate message passing based on learned relevance estimates. These architectures address in-

formation sharing and communication learning, but typically assume structurally symmetric agents whose interactions are organized around message passing alone. They do not distinguish evaluative and executory roles, integrate communication with reward mediation or reputation dynamics, or impose principled resource constraints on evaluative attention. Critically, most differentiable communication architectures require either a shared communication channel or a globally known interaction topology, both of which constitute implicit centralization. In the Observer–Actor framework, communication is not an isolated module but one component of a broader evaluative feedback structure in which selective attention, informational influence, and policy adaptation co-evolve from purely local interactions.

Role Differentiation and Heterogeneous Agents. Several approaches introduce behavioral specialization or role heterogeneity into MARL systems. ROMA **?** and RODE **?** learn latent role assignments that encourage agents to specialize behaviorally during coordination tasks, demonstrating that heterogeneous behavioral structure can improve coordination efficiency and scalability. VAST **?** and CoPO **?** introduce value decomposition structures that implicitly encourage role differentiation. However, role differentiation in these frameworks emerges primarily through auxiliary optimization objectives or latent representation learning rather than from an explicit account of evaluative structure. They generally do not address selective attention allocation under resource constraints, decentralized reputation dynamics, or communication-constrained evaluation. The Observer–Actor framework derives role asymmetry directly from the coordination problem itself: the evaluator–executor distinction is not an emergent artifact of an auxiliary objective but the architectural primitive from which all other coordination structures—attention hierarchies, influence asymmetries, informative representations—emerge endogenously.

Graph-Based and Attention-Based MARL. Graph neural network and attention-based approaches model coordination through structured interaction topologies and relational reasoning **??**. MAAC **?** introduces multi-head attention into a centralized critic to selectively weight contributions from different agents during value estimation. DGN **?** uses graph convolutional networks to propagate information across agent neighborhoods. More recent architectures such as G2ANet **?** and HAMA **?** extend these ideas to hierarchical and dynamic interaction graphs. These methods capture localized interactions and can scale to larger agent populations by exploiting sparse communication structure. However, they treat attention as an internal optimization mechanism within a critic or communication network rather than as an explicitly resource-constrained social process that shapes long-term influence dynamics. Attention weights in these architectures are computed per-step from global or semi-global information; they do not accumulate into persistent influence structures or emerge from purely local bilateral interactions. The Observer–Actor framework instead models attention allocation as a learnable object of coordination in its own right, with Observers dynamically allocating evaluative resources across Actors under explicit budget constraints, and influence hierarchies emerging endogenously from the accumulation of these local allocation decisions.

Social Influence and Social Reward Approaches. A related literature studies social influence and intrinsic social rewards in MARL **???**. **?** encourage agents to produce actions that causally influence the behavior of others, while **?** introduce influence-based communication learning. These approaches introduce socially mediated learning signals and have demonstrated that social influence rewards can improve coordination and communication emergence. However, they retain symmetric agent architectures and do not distinguish between agents that primarily evaluate and agents that primarily execute. Social reward is used as an auxiliary shaping signal rather than as the primary organizational principle of the system's architecture. The Observer–Actor framework is grounded in the companion theory of social reward economies developed in **?**, which conceptualizes social life as governed by dynamic systems of evaluative feedback in which agents adapt their behaviors to maximize socially mediated approval, recognition, and influence. The present framework provides a precise multi-agent reinforcement learning instantiation of this theory, making evaluative asymmetry the core organizational principle from which influence hierarchies and selective feedback structures emerge, rather than an auxiliary signal layered onto a symmetric architecture.

Hierarchical Reinforcement Learning. Hierarchical reinforcement learning (HRL) introduces asymmetric agent structures in which higher-level agents set goals or subgoals for lower-level agents **????**. HIRO **?** and Option-Critic **?** learn temporally extended action hierarchies that decompose complex tasks into subtasks. In multi-agent settings, methods such as HMARL **?** and feudal MARL **?** extend these ideas to populations of hierarchically organized agents. These approaches share with the Observer–Actor framework the intuition that coordination benefits from structured asymmetries between agents operating at different levels of abstraction. However, HRL architectures typically impose hierarchy top-down through architectural design rather than allowing it to emerge from local interactions, and the higher-level agent generally requires access to global or semi-global state to set meaningful subgoals. The influence hierarchies in the Observer–Actor framework emerge endogenously from accumulated bilateral feedback: no agent is designated as higher-level by design, and no agent requires non-local state information to participate in the feedback loop.

Mentor–Student and Advisor Architectures. Mentor–student, teacher–student, and advisor-based reinforcement learning frameworks introduce asymmetries between agents that provide guidance and agents that execute policies **????**. More recent work on learning from human feedback **??** and on policy distillation **??** similarly relies on asymmetric agent roles in which one agent provides evaluative signals that shape another agent's learning. These approaches share with the Observer–Actor framework the intuition that some agents may guide the learning of others through evaluative feedback. However, mentor–student architectures are typically externally imposed training structures rather than endogenous coordination mechanisms: the mentor is designated by design, the asymmetry does not emerge from local interactions, and the framework does not model reciprocal evaluative attention, evolving influence hierarchies, or decentralized communication learning among populations of interacting agents. The Observer–Actor framework extends beyond these formulations by treating the evaluator–executor asymmetry as an emergent, persistent property of the

deployed system rather than a fixed training scaffold.

Computational Sociology and Norm-Aligned Multi-Agent Systems. A smaller but directly relevant literature studies the use of sociological and social-scientific theory to inform multi-agent system design. Work on norm emergence and norm compliance in MAS **???** studies how behavioral norms can arise from local agent interactions without explicit programming. Social simulation approaches **??** use agent-based models to study emergent macro-level social phenomena from micro-level interaction rules. More recent work on value alignment and socially-aware MARL **???** studies coordination under social preferences and value constraints. The Observer–Actor framework is most directly grounded in the Socially Responsive Agent (SRA) model **?**, a formal computational model from sociology that operationalizes micro-level mechanisms of social feedback and behavioral adaptation identified by foundational sociological theory, including symbolic interactionism **?**, structuration theory **?**, and field theory **?**. Unlike prior work in this space, the Observer–Actor framework does not merely draw inspiration from social science but provides a precise algorithmic translation of the SRA model's feedback loop, with explicit update rules and formal convergence guarantees. The contribution is not sociological motivation layered onto a MARL architecture; it is the demonstration that the structural primitive sociology identifies—persistent bilateral evaluative asymmetry—is exactly the primitive needed to unify the components of the coordination problem within a single decentralized framework.

Summary. Table 1 summarizes the key properties of the Observer–Actor framework relative to existing approaches. Existing MARL approaches have developed powerful mechanisms for specific coordination subproblems: centralized critics address credit assignment, communication architectures address information exchange, role-based systems address specialization, attention mechanisms address selective information weighting, and social influence methods address socially mediated adaptation. Each of these mechanisms has been studied in isolation, optimized for benchmark performance, and evaluated on its own terms. No existing framework addresses all of them together, within a single mathematical structure, without centralized components, and with convergence guarantees that hold throughout. The Observer–Actor framework achieves this by deriving all coordination mechanisms from a single architectural primitive—persistent evaluator–executor asymmetry operating through bilateral dyadic interactions—that sociology identified as the structural feature making decentralized coordination tractable in systems far larger and more complex than any multi-agent benchmark.

Table 1: Comparison of the Observer–Actor framework with existing MARL approaches across key coordination properties. ✓ indicates the property is fully supported; ∘ indicates partial support; × indicates the property is not supported.

	Fully Decentralized	**Policy Learning**	**State Compression**	**Attention Allocation**	**Reward Mediation**	**Influence Hierarchies**	**Convergence Guarantees**	**Unified Framework**
CTDE (QMIX, MADDPG)	×	✓	×	×	×	×	∘	×
Diff. Communication	∘	✓	∘	∘	×	×	×	×
Role Differentiation	∘	✓	×	×	×	∘	×	×
Graph/Attention MARL	∘	✓	×	∘	×	×	×	×
Social Influence	✓	✓	×	×	∘	×	×	×
Hierarchical RL	×	✓	×	×	×	∘	∘	×
Mentor–Student	×	✓	×	×	∘	×	×	×
Observer–Actor	✓	✓	✓	✓	✓	✓	✓	✓

3 The Socially Responsive Agent

The Observer–Actor framework is grounded in the *Socially Responsive Agent (SRA)* model (**?**), which provides a formal account of how agents adapt their policies through repeated evaluative feedback from other agents. This section presents the SRA model, establishing the formal foundations on which the Observer–Actor framework is built.

The central mechanism is a feedback loop in which an agent's action elicits an evaluative signal from other agents in the system, which the acting agent uses to update its behavior:

$$action \;\rightarrow\; observation\,by\,others \;\rightarrow\; evaluative feedback \;\rightarrow\; behavior\,update.$$

This loop is the micro-level mechanism from which macro-level coordination structures—including behavioral norms, role differentiation, and influence asymmetries—emerge endogenously. The key property of the SRA model, and the one that motivates its use as a foundation for norm-aligned MAS, is that agents do not require explicit norm specifications or centralized reward engineering: structured collective behavior arises from the local, repeated exchange of evaluative feedback between agents.

3.1 Model

The SRA model is defined over dyadic interactions between two agents i and j, which serve as the basic building block of the framework (see Figure 1). Agent i selects an action, agent j observes and evaluates it, and returns a feedback signal that agent i uses to update its behavior. The model components are as follows.

1. *Contexts and Actions.* At each interaction, agents face a *context* $s \in \mathcal{S}$, representing the current state of the environment relevant to the interaction. In each context, agent i selects an action $a \in \mathcal{A}(s)$ whose outcome directly affects other agents in the system.

2. *Behavior.* Agent i's *behavior* π_i maps contexts to actions, where $\pi_i(s)$ denotes the action agent i takes in context s. The SRA model is concerned with how π_i evolves over time in response to feedback from other agents.

3. *Perceived Utility.* When agent i takes action a in context s, agent j evaluates the action according to a *perceived utility* $b_j(a, s) \in \mathbb{R}$, which can be positive or negative depending on how j is affected. This utility is private to agent j and not directly observable by agent i.

4. *Feedback Signal.* Agent j returns a *feedback signal* $f(b_j(a, s))$ to agent i, where $f(\cdot)$ is a monotone function of perceived utility: positive feedback encourages repetition of the action; negative feedback discourages it. Agent i treats this signal as a *reward*: empirical and neuroscientific evidence supports the view that evaluative social signals are processed by agents similarly to material reward signals (**???**).

5. *Behavior Update.* Agent i updates its behavior π_i to favor actions that generate higher cumulative feedback. Formally, the adaptive process tends to increase the accumulated reward across contexts:

$$\sum_{s \in \mathcal{S}} f(b_j(\pi_i(s), s)) \,.$$

   This is not an explicit optimization objective but an analytical description of the adaptive tendency: policies that consistently elicit positive feedback are reinforced, while those that elicit negative feedback are revised or abandoned.

6. *Feedback Neighborhood.* Agents do not treat feedback from all other agents equally. Each agent i maintains a *feedback neighborhood* $\mathcal{N}_i \subseteq \mathcal{G}$—a subset of agents whose evaluative signals receive primary weight in i's behavior updates. This models the observation that in multi-agent systems, agents are selectively responsive to feedback from relevant peers, teammates, or authorities, rather than the system as a whole. An agent may maintain distinct policies for distinct neighborhoods.

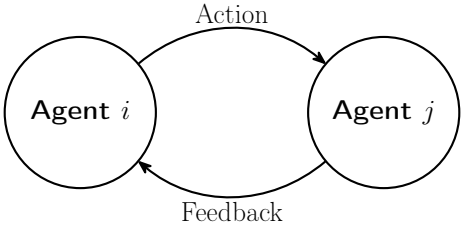


Figure 1: Basic dyadic interaction in the SRA model: agent i acts, agent j evaluates and returns a feedback signal.

3.2 Interaction Loop

Combining these components, the SRA interaction loop proceeds as follows (see Figure 2). Agent i faces context $s \in \mathcal{S}$ and selects action $a \in \mathcal{A}(s)$ according to its current behavior π_i. Agent $j \in \mathcal{N}_i$ observes the action, forms a perceived utility $b_j(a, s)$, and returns feedback $f(b_j(a, s))$. Agent i

treats this as a reward signal and updates π_i accordingly, increasing the probability of actions that elicit positive feedback and decreasing the probability of those that elicit negative feedback.

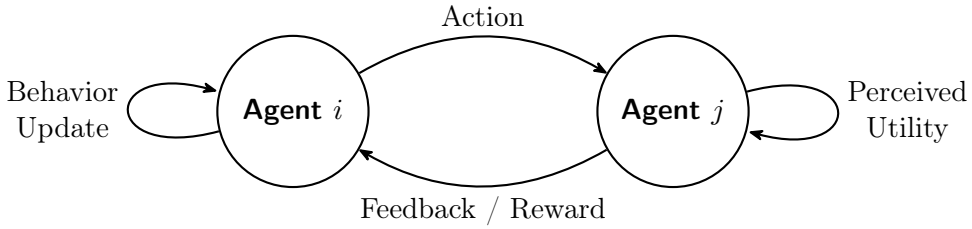


Figure 2: SRA interaction loop: actions generate feedback signals that drive behavior updates.

The adaptive process unfolds in two phases. In an initial *exploration* phase, agents trial different actions and observe the resulting feedback, building up a signal about which actions are positively received. In a subsequent *convergence* phase, agents concentrate probability on actions that consistently elicit positive feedback, settling into stable policies. Importantly, this stability is conditional: if the feedback structure of the environment shifts—due to changing agent populations or evolving evaluative criteria—agents return to exploration. The framework therefore supports both behavior stability and adaptation to environmental change.

Illustrative Example. Consider two agents, i and j, collaborating on a shared task. During a joint planning session (context s), agent i interrupts agent j mid-explanation to propose an alternative (action a). Agent j perceives this as counterproductive—$b_j(a, s) < 0$—and signals this through a reduction in cooperative engagement (feedback $f(b_j(a, s)) < 0$). Agent i registers this as a negative reward and reduces the probability of interrupting in future planning contexts. Repeated interactions of this kind, across all agents in the system, give rise to a shared behavioral norm against interruption in planning contexts—without any agent having been explicitly programmed with this norm.

3.3 From Dyadic Interactions to Group-Level Coordination

We now extend the dyadic SRA model to a group $\mathcal{G} = \{1, \ldots, N\}$ of agents, and show how repeated local interactions give rise to system-level coordination structures.

Each agent $i \in \mathcal{G}$ accumulates feedback across all agents in its neighborhood and all contexts. We define agent i's *reputation score*—its total accumulated feedback from the group—as:

$$R_i(\pi_i) \;=\; \sum_{j \in \mathcal{G} \setminus \{i\}} \sum_{s \in \mathcal{S}} f(b_j(\pi_i(s), s))\,.$$

Since $f(b_j(a, s)) \approx b_j(a, s)$, the reputation score approximates the total perceived utility agent i's behavior generates for other agents in the group. Agents update their policies to increase R_i, making the reputation score the effective optimization signal driving the system's joint dynamics.

A central result of the SRA framework (**????**) is that the joint behavior dynamics of a group of SRA agents converge to stable coordination structures. Specifically, three classes of macro-level structure emerge endogenously from local feedback interactions:

- *Behavioral norms:* convergent action profiles $\{\pi_i^*\}$ in which agents consistently select the same actions across shared contexts, without explicit norm specification (**?**).
- *Shared evaluation criteria:* alignment of the perceived utility functions $\{b_j(\cdot,\cdot)\}$ across agents, reflecting convergent standards for evaluating actions (**?**).
- *Influence hierarchies:* stable asymmetries in reputation scores $\{R_i\}$ across agents, giving rise to differentiated roles and authority structures (**?**).

These results establish the SRA model as a principled foundation for norm-aligned MAS: coordination structures are not engineered top-down but emerge from the bottom-up dynamics of feedback-driven behavior adaptation. The Observer–Actor framework, introduced in the next section, translates this mechanism into a concrete MAS architecture with explicit computational update rules and formal convergence guarantees.

3.4 From SRA to the Observer–Actor Framework

The SRA model serves as the conceptual foundation for the Observer–Actor framework: its bilateral feedback loop, reputation dynamics, and selective neighborhood structure directly motivate the architectural choices we make in designing the framework's update rules, role assignments, and convergence analysis. The Observer–Actor framework is, in a precise sense, a computational realization of the SRA model—one that translates its procedural description into algorithmic components with explicit update rules and formal guarantees.

At the same time, translating the SRA model into a full MAS architecture requires extending it in three directions that fall outside the scope of the original sociological framework.

1. *Following rates for social feedback.* In the SRA model, the feedback neighborhood $\mathcal{N}_i$ determines which agents' signals an actor attends to, but the intensity of that attention is not modeled as a learnable, resource-constrained quantity. In multi-agent settings with many Actors, an Observer cannot monitor all Actors equally: evaluative attention is a limited resource. We introduce *following rates*—learnable, budget-constrained weights that govern how strongly each Observer attends to each Actor's performance—as the algorithmic operationalization of this constraint. Observers allocate their following rates to maximize estimated Actor quality, giving rise endogenously to the influence asymmetries the SRA model predicts but does not mechanize.

2. *Instructive Feedback.* While traditional formulations of the Socially Responsive Agent (SRA) framework rely on evaluative feedback—where an observer merely relays a scalar metric of

perceived benefit or reward—we explicitly extend this paradigm to accommodate instructive feedback. Under this extended formulation, the feedback signal is not merely an ex-post evaluation of utility, but a directional vector in the agent's action or representation space. By communicating gradient information (i.e., specifying how an agent should perturb its behavior to maximize downstream utility), the observer provides explicit structural guidance—telling the actor how to adapt rather than just how well it performed.

Together, these three extensions preserve the decentralized, bilateral structure of the SRA model while adding the algorithmic machinery necessary to handle the informational realities of multi-agent environments. The resulting framework inherits the SRA model's convergence properties at the level of each extension, and—as we show in Section **??**—these properties compose when the extensions are combined.

3.5 Connection to Social Reward Economies

The Observer–Actor framework can be understood as a computational and algorithmic realization of the broader concept of social reward economies developed in companion work (Marbach and Jiang, 2026). In that framework, social life is conceptualized as governed by dynamic social reward economies—structured systems of social feedback in which individuals, as socially responsive agents, continually adapt their behaviors and interpretive schemas to maximize the social rewards (approval, recognition, status, and influence) they receive from others. These economies give rise to emergent macro-level phenomena such as group cohesion, polarization, resistance, and manipulation through recursive feedback processes.

The present Observer–Actor architecture provides a precise multi-agent reinforcement learning translation of this idea. By distinguishing between Observers (who evaluate actions, provide feedback, and allocate limited evaluative attention via learnable following rates) and Actors (who adapt their policies to maximize the socially mediated rewards they elicit), the framework makes explicit the asymmetric roles that underpin social reward economies. The following-rate mechanism (Section 4) directly operationalizes the selective attention and feedback weighting central to social reward economies, while reputation-like dynamics (via selection logits and accumulated feedback) mirror the emergence of influence hierarchies and opinion leaders.

Crucially, the Observer–Actor model preserves the decentralized, bilateral nature of social reward economies: coordination and structure arise endogenously from local interactions without requiring global state or centralized authority. In this sense, the framework not only grounds the sociological concept in rigorous algorithmic update rules and convergence analysis, but also demonstrates how the micro-dynamics of social reward pursuit can be formally studied and engineered in artificial multi-agent systems.

This connection highlights the bidirectional potential of the research program: the Observer–Actor framework supplies a concrete, simulatable model for exploring the dynamics of social reward economies, while the sociological perspective provides interpretive depth and conceptual grounding for the emergent behaviors observed in multi-agent learning.

4 Environmental Action with Evaluative Feedback

The SRA model establishes that feedback-driven policy adaptation between pairs of agents is the micro-level mechanism from which norm-aligned coordination emerges at scale. To operationalize this in a MAS setting, we need a computational framework that makes two roles explicit: an agent that *acts*, and an agent that *evaluates* that action and provides feedback. We call these agents *Actors* and *Observers*, respectively, and their interaction defines the Observer–Actor framework.

The Observer–Actor framework defines *interaction roles*, not agent identities. In any given interaction, the agent that produces the action under evaluation is designated as the *Actor*, and the agent that evaluates it and returns feedback is designated as the *Observer*. These roles are not fixed: the same agent may be an Actor in one interaction and an Observer in another, depending on the structure of the exchange.

The two role labels carry a specific meaning that is worth making explicit. *Actor* refers to the agent whose action is under evaluation—not necessarily the agent that interacts with the environment. In the informational-action case, for instance, it is the Observer who selects environmental actions; the Actor's role is to produce the compressed state representation that the Observer relies on to do so. The two cases introduced below should accordingly be understood as canonical building blocks of interaction—each isolating one type of evaluated action—rather than as descriptions of fixed agent types. Subsequent sections compose these building blocks into multi-agent systems in which agents simultaneously act, communicate, and evaluate, and in which roles emerge dynamically from interaction structure.

In the following, we introduce the framework in its simplest form: a single Observer and a single Actor. We consider two qualitatively distinct settings, each corresponding to a different type of action the Actor can perform.

In the first setting, the Actor performs an *environmental action with evaluative feedback*: it selects a state-dependent action in the environment. The Observer evaluates the reward generated by this action and reports it to the Actor as feedback. The Observer's role here is that of a reward mediator: it observes outcomes and provides evaluative feedback that drive the Actor's policy improvement.

In the second setting, the Actor performs an *informational action with instructive feedback*: it produces a compressed representation of the environment state and transmits it to the Observer as a message. The Actor's role here is to supply state information the Observer could not otherwise access; it is the Observer, not the Actor, who selects environmental actions, doing so on the basis of the transmitted message. The Observer evaluates the message by the quality of the decisions it supports, and returns feedback to the Actor indicating how useful the communicated state information was for downstream decision-making. Crucially, we extend the traditional SRA model here by framing this feedback as instructive feedback rather than a simple scalar valuation, explicitly instructing the Actor on how it should adapt its actions.

These two cases establish the core update rules and convergence properties that carry forward to the multi-agent settings of Sections **??**- 6.

4.1 Dyadic Interactions

We start with the case where the Actor performs an environmental action, and the Observer reports its reward obtained from the action.

For this, we use the following notation. The *State space* $\mathcal{S}$: is given by the set of possible environment states. At each time step t, the environment produces a state $s_t \in \mathcal{S}$.

Environmental Action by Actor. The *environmental action space* $\mathcal{A}$ is the set of environmental actions that are available to the actor.

The *policy* π_θ of the Actor is a parameterized stochastic policy governing environmental actions, where $\pi_\theta(a \mid s_t)$ conditions directly on the state.

Perceived Benefit by Observer. The perceived beneift of the Observer is given by the *reward function* $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ as a function of state s and action a. This reward mediated to the Actor by the Observer.

Interaction Loop At each time step t, the Observer–Actor interaction proceeds as follows:

1. The environment produces state $s_t \in \mathcal{S}$, fully observed by the Observer and Actor.
2. The Actor selects an action: $a_t \sim \pi_\theta(\cdot \mid s_t)$.
3. The environment produces reward $r_t = R(s_t, a_t)$, and transitions to state s_{t+1}.
4. The Observer observes the reward r_t and reports it to the Actor.
5. The Actor updates its parameter θ to maximize expected cumulative reward.

The separation between steps 2 and 4 —the Actor acts before observing its reward, which it receives only via the Observer—is central to the framework. It creates a cooperative dependency: the Actor requires the Observer to learn effectively, and the Observer's compression function is only useful insofar as it supports the Actor's policy improvement. This mutual dependency is the computational realization of the SRA feedback loop.

Objective. The Actor's goal is to find a policy π_θ that maximizes expected reward across the state distribution:

$$J(\theta) \;=\; \mathbb{E}_{s\sim\mathcal{D}}\Big[\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\big[R(s,a)\big]\Big]. \tag{1}$$

Policy Gradient. Applying the log-derivative trick to (1) yields the policy gradient:

$$\nabla_\theta J(\theta) \;=\; \mathbb{E}_{s\sim\mathcal{D},\, a\sim\pi_\theta(\cdot|s)}\Big[R(s,a)\,\nabla_\theta \log \pi_\theta(a \mid s)\Big]. \tag{2}$$

This is a standard REINFORCE-type gradient (**?**), adapted to the stochastic state setting. The expectation over $s \sim \mathcal{D}$ reflects the fact that the Actor must learn a policy that performs well across all states it may encounter, not just a single fixed state.

Update Rule. Using a sample (s_t, a_t, r_t) at time step t, the Actor updates its policy parameters via stochastic gradient ascent:

$$\theta_{t+1} \leftarrow \theta_t + \alpha\, r_t\, \nabla_\theta \log \pi_\theta(a_t \mid s_t), \tag{3}$$

where $\alpha > 0$ is the Actor's learning rate. The update reinforces action a_t in state s_t proportionally to the reward it generated: actions that produce high rewards in a given state become more probable; actions that produce low or negative rewards become less probable. Since the reward r_t is observed by the Observer and reported to the Actor, the Actor's learning depends entirely on the Observer faithfully relaying this signal. In this sense, even in the fully observable case, the Observer plays an active role in mediating the Actor's policy improvement.

No Observer update is performed in this setting, as the Observer function is not parameterized: its role is limited to reward reporting.

4.2 Following Rates

Observer i maintains a non-negative weight $\mu_{i,j}$ for each Actor A_j, representing the degree to which Observer O_i attends to Actor A_j's performance. These weights are subject to a budget constraint, modeling the intuition that an Observer's evaluative attention is a limited resource. Actor policies remain fixed in this setting.

If

1. actor j is active at time t and
2. performs action a_t,

and observer i follows actor j with rate $\mu_{i,j}$, then observer i provides actor j reward information, i.e. reports its observered reward $R_i(s_t, a_t)$, with probabilty $\theta(\mu_{i,j})$, where

$$\theta(\mu) = 1 - e^{-c\mu},$$

where $c > 0$ is a positive constant.

Following Rates. Observer O_i maintains a *following-rate vector* $\boldsymbol{\mu}_i = (\mu_{i,1}, \ldots, \mu_{i,M})$ subject to the budget constraint:

$$\sum_{j=1}^{N} \mu_{i,j} \;\le\; B, \qquad \mu_{i,j} \;\ge\; 0, \tag{4}$$

where $B > 0$ is a fixed budget shared across all Observers. At each time step t, a single Actor A_{j_t} is selected to interact with the environment—for example, uniformly at random or according to an exogenous schedule. Actor A_{j_t} executes action $a_t \sim \pi^{(j_t)}(\cdot \mid s_t)$, generating reward $R_i(s_t, a_t)$ for Observer i.

Actor Selection. At each time step t, a single actor is selected to perform an action. The probability that Actor j is selected is p_j.

Reward function $R_i : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. The reward $R_i(s, a)$ is observed by the Observer i if action a is performed in state s.

Actor Quality Estimation. Observer O_i maintains a running estimate $\hat{U}_{i,j}$ of the average reward generated by Actor A_j:

$$U_{i,j} \;=\; \mathbb{E}_{s,a}\Big[R_i(s, a) \;\Big|\; a \sim \pi^{(j)}(\cdot \mid s)\Big]. \tag{5}$$

When Actor j acts at step t, Observer O_i updates its estimate using a stochastic approximation step:

$$\hat{U}_{i,j_t,t+1} \;=\; \hat{U}_{i,j_t,t} + \alpha_U\big(R_i(s_t, a_t) - \hat{U}_{i,j_t,t}\big), \tag{6}$$

while estimates for inactive Actors are held fixed: $\hat{U}_{i,j,t+1} = \hat{U}_{i,j,t}$ for $j \neq j_t$.

Observer Objective. Given estimates $\{\hat{U}_{i,j}\}$, each Observer O_i allocates its following rates to maximize estimated total reward subject to the budget constraint:

$$\max_{\boldsymbol{\mu}_i} \sum_{j=1}^{N} \theta(\mu_{i,j})\, \hat{U}_{i,j} \qquad \text{s.t.} \qquad \sum_{j=1}^{N} \mu_{i,j} \;\le\; B,\;\; \mu_{i,j} \;\ge\; 0. \tag{7}$$

Following-Rate Update. Each Observer updates its following-rate vector using projected gradient ascent on the objective (7). An unconstrained gradient step is first applied:

$$\hat{\mu}_{i,j}(t+1) \;=\; \mu_{i,j}(t) + \alpha_\mu\, \theta'(\mu_{i,j}(t))\hat{U}_{i,j,t}, \qquad j = 1, \ldots, N. \tag{8}$$

Since this step may violate the nonnegativity and budget constraints in (4), the result is projected back onto the feasible simplex:

$$\boldsymbol{\mu}_i(t+1) \;=\; \Pi_{\{\boldsymbol{\mu} \ge 0,\, \sum_j \mu_j \le B\}}\big(\hat{\boldsymbol{\mu}}_i(t+1)\big). \tag{9}$$

The projection (9) has a closed-form solution via the simplex projection algorithm (**?**), making each update computationally efficient regardless of M. No Actor update is performed in this setting.

Two-Timescale Dynamics. The update (6) operates on a different timescale compared with the other udpates. That is, we want the utilities updates to (almost) converge to use them in the update (8). To achieve this, we let the learning rate α_U to be larger compared with the learning rate α_μ.

Experiments:

- Show that observer follows high-utility actors with a higer rate.

4.3 Multiple Observers and Multiple Actors

We now extend the dyadic reward-information setting of Section 4.1 to groups of Observers and Actors. In this section, all Actors have direct access to the full environment state, so no state compression or message passing is required. The central question is how Observers learn to allocate attention across multiple Actors—and, in the most general case, how Actors simultaneously learn to improve their policies in response to Observer-mediated reward signals.

We develop this in three steps. Section **??** considers the case in which each Observer selects a single Actor at each step via a learnable discrete distribution, with Actor policies fixed. Section 4.2 replaces discrete selection with a continuous following-rate allocation under a budget constraint, introducing a critic-style evaluation mechanism. Section **??** allows Actor policies to adapt simultaneously with Observer selection, yielding the fully joint learning setting.

Throughout, the environment state s_t is sampled i.i.d. from $\mathcal{D}$ at each time step t. There are N Actors $A_1, \ldots, A_N$ and N Observers $O_1, \ldots, O_N$.

At each time step, a single actor is selected with Actor j being selected with probability p_j.

- **State Dynamics:** At each time step t, the environment produces a state $s_t \sim \mathcal{D}$.
- **Actor Set:** N Actors, $A_1, \ldots, A_N$, each with a fixed policy $\pi^{(j)}(a|s_t)$.
- **Observer Set:** N Observers, $O_1, \ldots, O_N$.
- **Following Rates:** Each Observer O_i maintains nonnegative following rates $\boldsymbol{\mu}_i = (\mu_{i,1}, \ldots, \mu_{i,N})$, representing how strongly Observer O_i follows each Actor.
- **Budget Constraint:** Each Observer has a fixed budget B such that

$$\sum_{j=1}^{M} \mu_{i,j} \leq B, \quad \mu_{i,j} \geq 0. \tag{10}$$

- **Actor Activation:** At each time step t, a single Actor j_t is selected to act.
- **Environment Interaction:** The active Actor executes $a_t \sim \pi^{(j_t)}(\cdot|s_t)$, producing a reward $R_i(s_t, a_t)$ for Observer i.

In this setting, both Observers and Actors learn simultaneously.

Setup. Each Actor A_j now has a learnable, parameterized policy $\pi_{\theta_j}(a \mid s_t)$. Each Observer O_i maintains following rates μ_i. Actor j is selected at time t with probability p_j.

Actor's Observed Reward. From Actor j's perspective, the reward that it observes is given by

$$\mathbb{E}_{s,a}\left[p_j \sum_{i=1}^{N} \theta(\mu_{i,j}) R_i(s,a) \;\middle|\; a \sim \pi^{(j)}(\cdot \mid s) \right]. \tag{11}$$

This captures both the intrinsic quality of an Actor's policy and the degree to which its actions are monitored by Observers.

Update Rules. Two simultaneous updates are performed at each step t.
Actor policy update. If Actor j is selected at time t (recall that Actor j is selected with probability p_j), and performs action a_t in state s_t; and if Observer i follows Actor j at time t (recall that Observer i follows actor j with probability $\theta(\mu_{ij}(t))$) and informs j on its recived reward $R_i(a_t, s_t)$, then Actor j updates its policy using

$$\theta_i(t+1) \leftarrow \theta_i(t) + \alpha_\theta \left[\sum_{i=1}^{N} \theta(\mu_{i,j}) R_i(a_t, s_t) \right] \nabla_{\theta_i} \log \pi_{\theta_i(t)}(a_t \mid s_t). \tag{12}$$

Observer following-rates update. **Same as in the previous subsection**.

Two-Timescale Dynamics. The update (6) operates on a different timescale compared with the other udpates. That is, we want the utilities updates to (almost) converge to use them in the other updates. To achieve this, we let the learning rate α_U to be larger compared with the other learning rates.

5 Informational Actions with Instructive Feedback

We now consider the case where the Actor performs an informational action. At each time step t, the environment produces a full state $s_t \in \mathcal{S}$, which is not directly available to all agents. Instead, information must be transmitted and evaluated through a structured interaction between two roles: an *Actor*, whose primary function is to produce compressed state information, and an *Observer*, whose role is to evaluate this information and provide feedback.

We use the following notation. The *state information message space* $\mathcal{M}$ is the the space of messages through which the Actor communicates state information to the Observer.

5.1 Dyadic Interactions

Informational Action by Actor. The informational actions is defined by a *state information function* $f_\phi : \mathcal{S} \to \mathcal{M}$ which is a learnable mapping that produces compressed state representations $m_t = f_\phi(s_t)$.

Perceived Benefit by Observer. The perceived benefit is given by the expected reward achieved by the Observer when acting under a policy conditioned on the transmitted message. In particular, the message $m = f_\phi(s)$ is made available to the Observer's policy π_θ, and the utility is defined as the resulting expected return:

$$U(m, s) \;:=\; \mathbb{E}_{a\sim\pi_\theta(\cdot|o,m)}\big[R(s, a)\big],$$

where the expectation is taken over actions selected by the Observer in state s, conditioned on its local observation o and the communicated message m.

Feedback by Observer as Evaluative Signal. The Observer provides an evaluative signal that reflects the downstream effectiveness of the communicated state information. Unlike the environmental-action setting, where feedback is a scalar reward $r_t \in \mathbb{R}$, the informational-action setting induces a structured signal over the message space that reflects how changes in the communicated representation influence downstream decision quality.

We construct an influence-based feedback signal in message space,

$$\delta_t := r_t \cdot \nabla_m \log \pi_\theta(a_t \mid o_t, m_t),$$

which measures how infinitesimal perturbations of the message would have affected the probability of the action that generated the observed reward. Consequently, $\delta_t \in \mathbb{R}^{|\mathcal{M}|}$ defines a directional signal indicating how the message should be modified to improve expected downstream return.

This formulation shifts the role of feedback from scalar outcome evaluation to directional guidance in representation space.

Interaction Loop. At each time step t, the Actor–Observer interaction proceeds as follows:

1. The environment produces a state $s_t \in \mathcal{S}$.
2. The Actor performs an informational action by producing a compressed state representation

   $$m_t = f_\phi(s_t) \in \mathcal{M},$$

   where f_ϕ is the Actor's state-encoding policy.
3. The Observer selects an action $a_t \sim \pi_\theta(\cdot \mid o_t, m_t)$, observes reward $r_t = R(s_t, a_t)$, and computes the feedback signal $\delta_t = r_t \cdot \nabla_m \log \pi_\theta(a_t \mid o_t, m_t)$.

4. The Observer transmits δ_t to the Actor.

5. The Actor updates its parameters θ to improve the quality of future state representations.

Actor Update. The Actor updates its encoding function using feedback from the Observer:

$$\phi_{t+1} \leftarrow \phi_t + \alpha_\phi \, r_t \cdot \underbrace{\nabla_m \log \pi_\theta(a_t \mid o_t, m_t)}_{\text{Observer's sensitivity to message}} \cdot \underbrace{\nabla_\phi f_\phi(s_t)}_{\text{compression Jacobian}} . \tag{13}$$

Note that the Observer does not evaluate the message directly in isolation, but instead assesses it through its impact on downstream decision quality. In particular, the usefulness of a message is defined by how much it improves the Observer's ability to select high-reward actions.

We therefore define the effective feedback signal as

$$\delta_t := r_t \cdot \nabla_m \log \pi_\theta(a_t \mid o_t, m_t),$$

which captures the marginal contribution of the communicated state information to realized reward via its influence on the Observer's policy.

The term

$$\nabla_\phi f_\phi(s_t)$$

is computed locally by the Actor and specifies how changes in the encoder parameters affect the produced message.

The two terms in this update have a clear interpretation. The first, $\nabla_m \log \pi_\theta(a_t \mid o_t, m_t)$, measures how sensitive the Observer's action probabilities are to changes in the message: it identifies which dimensions of m_t the Observer is currently relying on for its decisions. The second, $\nabla_\phi f_\phi(s_t)$, is the Jacobian of the compression function with respect to its parameters: it determines how ϕ should change to alter the message in the desired direction. Together, these terms guide the Actor toward compressions that highlight the features of s_t most relevant to the Observer's current policy, causing the Observer's messages to become increasingly informative over the course of learning.

Information Encapsulation and Dimensional Bottlenecks in Dyadic Sharing. A common critique of frameworks that propagate gradients through communication channels is that they introduce hidden centralized training dependencies or violate the decentralized integrity of individual entities. In our formulation, the update rule for the compression parameters ϕ_{k_t} in Equation (**??**) resolves these concerns through two structural properties: topological isolation and dimensional bottlenecking.

First, the gradient flow required by Equation (**??**) is strictly *dyadic* (bilateral) rather than global.

Second, our framework enforces a rigorous *dimensional isolation bottleneck*. In typical differentiable communication architectures, backpropagation metrics scale with the size of large internal network representations or require unconstrained parameter sharing. In contrast, our differentiation occurs strictly with respect to the compressed message space $\mathbb{R}^d$, where $d \ll \dim(\mathcal{S})$ and

$d \ll \dim(\Theta)$. Consequently, the vector passed backward,

$$\nabla_m \log \pi_\theta(a_t \mid m_t) \in \mathbb{R}^d, \tag{14}$$

imposes a communication bandwidth overhead that is identical to the forward pass. This bandwidth requirement scales linearly with the compression dimension d and remains completely invariant to the internal complexity, depth, or parameter size of the Actor's neural network π_θ.

Crucially, this structural bottleneck maintains strict *agent encapsulation*. To execute the backward pass, the Actor functions as a "grey box" rather than a transparent one. The Actor never exposes its private local observations o_t, its internal policy weights θ, or its learning objectives to the Observer. It merely transmits an external marginal sensitivity vector across the bilateral interface, communicating how a slight perturbation in the received message would have shifted the log-probability of its executed action. This formulation mathematically mirrors the sociological intuition of the Socially Responsive Agent framework: an advisee does not expose their entire neural architecture to a mentor; they simply provide low-dimensional, localized feedback indicating the behavioral utility of the advice received.

Interpretation as SRA Dynamics. This formulation can be interpreted as a specialization of the SRA framework in which the Actor's action is informational rather than environmental. The Actor generates a representation of the state, the Observer evaluates this representation, and feedback propagates back to refine the Actor's encoding strategy.

Over time, this feedback loop induces progressively more informative state representations, as the Actor learns to align its encoding with the Observer's evaluative criteria. In this sense, coordination emerges not through action in the environment, but through the iterative refinement of shared state representations under evaluative feedback.

5.2 Actor Selection for State Information

We now extend the informational-action paradigm to a setting where a single downstream Observer selects among a group of competing state-compressing Actors.

Throughout this section, the environment state s_t is sampled i.i.d. from a distribution $\mathcal{D}$ at each time step t. There are N distinct Actors $\{A_i\}_{i=1}^N$, each maintaining a localized, learnable parameter set ϕ_i for its compression function $f_{\phi_i} : \mathcal{S} \to \mathcal{M}$ that maps the full state to a message $m_t^i = f_{\phi_i}(s_t)$. There is a single, centralized Observer O that maintains a fixed, pre-specified policy $\pi_\theta : \mathcal{M} \to \Delta(\mathcal{A})$ that maps a compressed message to an action distribution over environmental actions.

In the following we use the notation

$$r(s, a) = R(s, a)$$

and

$$r_t = R(s_t, a_t),$$

representing the environmental reward evaluated by the Observer based on the downstream action taken.

We assume that the Observer's policy π_θ is fixed and pre-specified. The learning problem is two-fold: the individual Actors learn to optimize their compression parameters $\{\phi_i\}$, while the Observer faces an *Actor-selection problem*: based on the environmental rewards generated by executing actions under different message streams, the Observer must learn which Actor's compression pathway to select and rely upon.

Setup. The Observer maintains a vector of learnable *selection logits* $\mathbf{w} = (w_1, \ldots, w_N) \in \mathbb{R}^N$, from which Actor-selection probabilities are derived via the softmax transformation:

$$\lambda_i(\mathbf{w}) \;=\; \frac{e^{w_i}}{\sum_{j=1}^N e^{w_j}}, \qquad i = 1, \ldots, N. \tag{15}$$

At each time step t, the Observer samples a selected Actor index $k_t \sim \lambda(\mathbf{w})$. The selected Actor A_{k_t} observes the state s_t, computes its message $m_t = f_{\phi_{k_t}}(s_t)$, and transmits it to the Observer. The Observer executes an environmental action $a_t \sim \pi_\theta(\cdot \mid m_t)$, receives the environmental reward $r_t = R(s_t, a_t)$, and uses this scalar signal to update its selection logits $\mathbf{w}$ and provide feedback to the selected Actor.

Update Rules. Two parameter sets are updated simultaneously from the sample (s_t, k_t, a_t, r_t):

1. **Observer's Selection Logits:** Substituting the softmax gradient into a stochastic gradient

ascent step, the Observer updates its attention profile:

$$w_{k_t,t+1} \leftarrow w_{k_t,t} + \alpha_w \, r_t \, (1 - \lambda_{k_t}), \tag{16}$$

$$w_{j,t+1} \leftarrow w_{j,t} - \alpha_w \, r_t \, \lambda_j, \qquad j \neq k_t. \tag{17}$$

2. **Selected Actor's Compression Parameters:** The chosen Actor A_{k_t} updates its encoding parameters using the evaluative feedback relayed by the Observer:

$$\phi_{k_t,t+1} \leftarrow \phi_{k_t,t} + \alpha_\phi \, r_t \cdot \nabla_{\phi_{k_t}} \log \pi_\theta(a_t \mid f_{\phi_{k_t}}(s_t)) \tag{18}$$

   The compression parameters of unselected Actors $j \neq k_t$ remain unchanged.

Remark. Equations (16) and (17) implement a form of *informational reputation*: Actors whose compression models consistently retain the specific state features vital for the Observer's policy to score high rewards accumulate higher logits and are selected with greater frequency. This is a direct computational analogue to the emergence of specialized information authorities in the SRA framework (Section 3.3), where agents who act as high-fidelity "sensors" or interpreters for the group expand their structural influence over collective decision-making.

Experiments:

- Show that the Observer's selection mechanism successfully concentrates its selection probability λ on the specific Actors whose compression mapping f_{ϕ_i} minimizes information loss on task-relevant state dimensions.

5.3 Multiple Actors and Multiple Observers

We now consider the complete, decentralized dyadic setting in which all learnable components adapt simultaneously across populations of both agents. In this framework, multiple Observers independently learn which localized Actor-compressors to monitor, while simultaneously optimizing their own decision policies. Concurrently, individual Actors adapt their state-compression parameters based on the feedback streams generated by the specific Observers that select them. This architecture fully computationalizes the SRA group-level feedback loop: behavioral policies, selective attention profiles, and localized informational representations co-evolve organically across the network.

Setup. There are M distinct Actors $\{A_j\}_{j=1}^{M}$ and N distinct Observers $\{O_i\}_{i=1}^{N}$. Each Actor A_j maintains independent compression parameters ϕ_j for its mapping $f_{\phi_j} : \mathcal{S} \to \mathcal{M}$. Each Observer O_i maintains its own behavioral policy parameters θ_i alongside a private vector of learnable *selection logits* $\mathbf{w}_i = (w_{i,1}, \ldots, w_{i,M}) \in \mathbb{R}^M$, from which its channel-attention probabilities $\lambda_i(\mathbf{w}_i)$ are derived:

$$\lambda_{i,j}(\mathbf{w}_i) \;=\; \frac{e^{w_{i,j}}}{\sum_{l=1}^{M} e^{w_{i,l}}}, \qquad j = 1, \ldots, M. \tag{19}$$

At each time step t, the global state $s_t \sim \mathcal{D}$ is sampled. Every Observer O_i independently samples an Actor index $k_i(t) \sim \lambda_i(\mathbf{w}_i)$, queries that specific Actor for its compressed message $m_t^{k_i} = f_{\phi_{k_i}}(s_t)$, executes a localized action $a_t^i \sim \pi_{\theta_i}(\cdot \mid m_t^{k_i})$, and experiences an individual environmental reward $r_t^i = R_i(s_t, a_t^i)$.

Update Rules. At each step t, updates are performed simultaneously across all interacting agent parameter spaces based on the local interaction tuple $(s_t, k_i(t), a_t^i, r_t^i)$:
Observer policy updates. Each Observer O_i updates its independent decision parameters using its local policy gradient and experienced reward, conditioning directly on the message stream provided by its chosen Actor:

$$\theta_{i,t+1} \;\leftarrow\; \theta_{i,t} + \alpha_\theta \, r_t^i \, \nabla_{\theta_i} \log \pi_{\theta_i}(a_t^i \mid m_t^{k_i}). \tag{20}$$

Observer selection logit updates. Each Observer O_i updates its own attention logits to optimize future channel selection, shifting weight toward the specific Actor-compressor that yielded higher utility:

$$w_{i,k_i(t),t+1} \;\leftarrow\; w_{i,k_i(t),t} + \alpha_w \, r_t^i \, (1 - \lambda_{i,k_i(t)}), \tag{21}$$

$$w_{i,j,t+1} \;\leftarrow\; w_{i,j,t} - \alpha_w \, r_t^i \, \lambda_{i,j}, \qquad j \neq k_i(t). \tag{22}$$

Actor compression updates. Because an Actor A_j may be simultaneously selected and queried by a subset of different Observers at step t, its compression parameters ϕ_j adapt by aggregating the

incoming instructive feedback vectors from all Observers that listened to it:

$$\phi_{j,t+1} \;\leftarrow\; \phi_{j,t} + \alpha_{\phi} \sum_{i:k_i(t)=j} \left[\delta_{i,t}^{T} \nabla_{\phi_j} f_{\phi_j}(s_t)\right], \tag{23}$$

where the instructive feedback vector $\delta_{i,t}$ transmitted from Observer O_i to its chosen Actor is defined cleanly as:

$$\delta_{i,t} \;:=\; r_t^i \cdot \nabla_m \log \pi_{\theta_i}(a_t^i \mid m)\big|_{m=m_t^j}. \tag{24}$$

If an Actor j is not selected by any Observer at time step t, its parameters remain unchanged.

6 Group-Level Interactions with Environmental and Informational Actions

We now consider the fully generalized group setting where the rigid structural distinction between fixed "Actor" and "Observer" identities is entirely dissolved. Instead, the system consists of a homogeneous population of N agents, $\{1, \ldots, N\}$, where each agent simultaneously executes both informational actions (compressing and broadcasting state observations) and environmental actions (monitoring peers, allocating attention, and making physical decisions).

This setting realizes the complete, symmetric SRA feedback loop: every agent acts as an informational authority to its peers while concurrently relying on those same peers as structural mentors. Long-term coordination, selective attention networks, and localized state representations emerge organically from reciprocal, bilateral feedback.

Setup. At each time step t, the global environment state $s_t \sim \mathcal{D}$ is sampled. Every agent $i \in \{1, \ldots, N\}$ observes the state and utilizes its private compression function $f_{\phi_i} : \mathcal{S} \to \mathcal{M}$, parameterized by ϕ_i, to generate a localized message stream $m_t^i = f_{\phi_i}(s_t)$.

Concurrently, each agent i faces an attention-allocation problem to determine which peer's information channel to follow. Agent i maintains a private vector of learnable *following logits* $\mathbf{w}_i = (w_{i,1}, \ldots, w_{i,N}) \in \mathbb{R}^N$. From these logits, continuous attention allocation probabilities $\lambda_{i,j}(\mathbf{w}_i)$ (the probability that agent i follows peer j) are derived via the softmax transformation:

$$\lambda_{i,j}(\mathbf{w}_i) \;=\; \frac{e^{w_{i,j}}}{\sum_{l=1}^{N} e^{w_{i,l}}}, \qquad j = 1, \ldots, N. \tag{25}$$

Each agent i independently samples a peer index $k_i(t) \sim \lambda_i(\mathbf{w}_i)$, retrieves the compressed message $m_t^{k_i} = f_{\phi_{k_i}}(s_t)$ generated by that peer, and executes a physical environmental action $a_t^i \sim \pi_{\theta_i}(\cdot \mid m_t^{k_i})$ governed by its learnable policy parameters θ_i.

As a consequence of the joint physical actions $\mathbf{a}_t = (a_t^1, \ldots, a_t^N)$, agent i experiences an individual environmental reward $R_i(s_t, \mathbf{a}_t)$.

The Weighted Reward Objective. Because information channels are decentralized and non-exclusive, a single compressing agent j may be simultaneously monitored by a changing subset of listening peers. To optimize its compression parameters ϕ_j, agent j must evaluate its informational utility using a mediated reward metric. This objective aggregates the rewards of all peers in the environment, weighted dynamically by the probability that each peer actively follows agent j:

$$r_j(s, \mathbf{a}) \;=\; \sum_{i=1}^{N} \lambda_{i,j}(\mathbf{w}_i) R_i(s, \mathbf{a}). \tag{26}$$

At time step t, the realized sample reward observed by agent j for updating its representation is:

$$r_{j,t} \;=\; \sum_{i=1}^{N} \lambda_{i,j}(\mathbf{w}_i(t)) R_i(s_t, \mathbf{a}_t). \tag{27}$$

Update Rules. Four parameter sets co-evolve synchronously at each time step t across the interacting population. For every agent $i \in \{1, \ldots, N\}$:

1. **Environmental Policy Update:** Agent i updates its decision parameters θ_i using the localized policy gradient, treating the message $m_t^{k_i}$ received from its chosen peer as the local state input:

$$\theta_{i,t+1} \;\leftarrow\; \theta_{i,t} + \alpha_\theta \, R_i(s_t, \mathbf{a}_t) \, \nabla_{\theta_i} \log \pi_{\theta_i}(a_t^i \mid m_t^{k_i}). \tag{28}$$

2. **Attention Logit Update:** Agent i modifies its following logits $\mathbf{w}_i$ to adjust its future network reliance, shifting attention mass toward the peer whose compressed perspective maximized agent i's local environmental utility:

$$w_{i,k_i(t),t+1} \;\leftarrow\; w_{i,k_i(t),t} + \alpha_w \, R_i(s_t, \mathbf{a}_t) \, (1 - \lambda_{i,k_i(t)}), \tag{29}$$

$$w_{i,l,t+1} \;\leftarrow\; w_{i,l,t} - \alpha_w \, R_i(s_t, \mathbf{a}_t) \, \lambda_{i,l}, \qquad l \neq k_i(t). \tag{30}$$

3. **Informational Compression Update:** As a broadcaster of state information, agent i updates its localized compression parameters ϕ_i by aggregating the instructive feedback vectors transmitted back from the subset of peers that chose to listen to it at step t:

$$\phi_{i,t+1} \;\leftarrow\; \phi_{i,t} + \alpha_\phi \sum_{l : k_l(t) = i} \left[\delta_{l,t}^T \nabla_{\phi_i} f_{\phi_i}(s_t) \right], \tag{31}$$

   where the instructive feedback vector $\delta_{l,t}$ encapsulates the downstream parameter sensitivity of the listening peer l, defined using the globally mediated reward metric:

$$\delta_{l,t} \;:=\; r_{i,t} \cdot \nabla_m \log \pi_{\theta_l}(a_t^l \mid m) \big| design_{m = m_t^i}. \tag{32}$$

   If no peers choose to follow agent i at time step t, its compression parameters remain static ($\phi_{i,t+1} \leftarrow \phi_{i,t}$).

Two-Timescale Dynamics. The concurrent optimization of environmental execution (θ), selective attention ($\mathbf{w}$), and state abstraction (ϕ) introduces severe non-stationarity if all components adapt at identical rates. To decouple these interactions and guarantee stable convergence, the framework implements a *two-timescale stochastic approximation profile.*

The attention allocation profiles $\lambda_{i,j}(\mathbf{w}_i)$ and the policy trajectories π_{θ_i} operate on a fast learning timescale, whereas the underlying state compression parameters ϕ_i evolve on a slower learning

timescale. Mathematically, this is enforced by structuring the learning rate sequences such that:

$$\lim_{t\to\infty} \frac{\alpha_\phi}{\alpha_\theta} = 0, \qquad \text{and} \qquad \lim_{t\to\infty} \frac{\alpha_\phi}{\alpha_w} = 0. \tag{33}$$

Under this layout, the network's attention profiles and decision policies rapidly reach a quasi-stationary equilibrium relative to the slow-moving message features. The compression mappings $\{\phi_i\}$ can then treat the fast-adapting layers as an effectively converged baseline. This stabilizing architecture prevents feedback-loop explosions and ensures that representation updates track long-term, structural coordination utility rather than transient environmental noise.

7 Conclusion

The Observer–Actor framework makes a single, precise claim: persistent bilateral evaluator–executor asymmetry is the structural primitive from which decentralized multi-agent coordination can be derived. Not assembled—derived. Communication, selective attention, state compression, reward mediation, role differentiation, and influence hierarchies are not independent modules that happen to coexist within a single architecture. They are interlocking consequences of one bilateral interaction primitive, and each falls out of that primitive without auxiliary objectives, bespoke analysis, or centralized scaffolding at any stage of training or execution.

This distinction—derivation rather than assembly—is the source of every property that distinguishes the framework from existing approaches, and it is worth making explicit why.

Comprehensiveness without centralization. Prior frameworks that achieve breadth do so by adding components, and added components introduce centralized dependencies. A centralized critic handles credit assignment; a global communication channel handles information exchange; an external role-assignment procedure handles specialization. Each component solves one coordination subproblem while creating a bottleneck that limits scalability and requires either global state or centralized training infrastructure. The Observer–Actor framework achieves comparable breadth without any of these dependencies because its components are not added—they emerge. The evaluative feedback loop that drives policy learning is the same loop that shapes compression functions, that allocates following rates, and that gives rise to influence hierarchies. There is nothing to centralize because there is no separate module to coordinate.

Convergence guarantees that compose. Formal convergence guarantees for individual MARL components are available in the literature. Guarantees that hold when those components interact simultaneously are not. The Observer–Actor framework provides both, and the reason is structural rather than mathematical: the SRA feedback loop naturally induces a two-timescale update structure in which compression parameters evolve on a slower timescale than policies and attention profiles. This timescale separation is not imposed to make proofs tractable—it is a property of the bilateral primitive itself. The convergence results therefore do not require case-by-case mathematical engineering; they are consequences of choosing the right architectural primitive.

Scalability as an architectural property. The framework's scaling advantage over CTDE methods is not incidental—it is a direct consequence of what a centralized critic is required to do. A centralized critic must observe global state, aggregate rewards across all agents, and estimate value functions that condition on a joint action space that grows exponentially with the number of agents. This bottleneck is architectural: it cannot be engineered away, because it reflects the information requirements of centralized evaluation itself. In the Observer–Actor framework, each agent's computational and communication burden is bounded by the size of its local interaction neighborhood, regardless of population size. Influence hierarchies, selective attention networks,

and shared behavioral norms emerge from the accumulation of bilateral interactions; no agent ever requires information it could not obtain through dyadic exchange. As the population grows, coordination capacity scales by adding bilateral pairs rather than by expanding any centralized component. This architectural property mirrors how large-scale human social coordination actually works—and the framework is, to our knowledge, the first MARL architecture to instantiate it formally.

The role of sociology. The sociological grounding of this framework is not motivational. It is the reason the framework exists in the form it does. The technical ingredients required to construct the Observer–Actor framework—policy gradients, two-timescale stochastic approximation, projected gradient ascent, differentiable communication—have been available to the MARL community for years. The framework does not exist in the prior literature not because the mathematics was unavailable, but because the conceptual primitive was not in hand. Sociology identified that primitive: the persistent, bilateral asymmetry between agents who evaluate and agents who act is the structural feature that makes decentralized coordination tractable in human social systems operating at scales far beyond any multi-agent benchmark. Translating that insight into a formal algorithmic architecture is what the Observer–Actor framework accomplishes. The absence of this framework in the existing literature despite the availability of all necessary mathematical tools is itself evidence that the conceptual insight precedes the architecture—and that the insight is a sociological one.

7.1 Extension to MDP

Throughout the paper we have adopted an i.i.d. state sampling assumption in place of a full Markov decision process formulation. This is a deliberate presentational choice: the i.i.d. setting isolates the coordination mechanisms of interest—feedback-driven policy adaptation, state compression, and attention allocation—from the additional complexity of temporal credit assignment, keeping the notation light and the framework's conceptual structure visible.

The extension to the MDP setting requires no structural modification to the framework. Each agent learns a *local* Q-function over the state–action pairs induced by its own bilateral interactions: the Q-value $Q_i(s, a)$ estimates the expected cumulative reward that agent i obtains from taking action a in state s, computed entirely from locally observable information without access to global state or other agents' rewards. Immediate rewards are replaced by these local Q-values throughout, and each Actor updates its policy using estimated Q-values as action-value signals. The Critic estimates the local Q-function for each dyadic interaction using standard temporal-difference machinery. Under standard assumptions from the Actor–Critic literature—step-size schedules satisfying the Robbins–Monro conditions, timescale separation between Actor and Critic updates, sufficient exploration, and regularity conditions on function approximation **?**—the resulting learning dynamics inherit convergence guarantees for each dyadic component.

The key point is that every dyadic interaction remains bilateral and local throughout this ex-

tension. Extensions that would require global redesign in centralized frameworks—because the centralized critic must be reconstructed to accommodate temporal dependencies—reduce here to local substitutions within each bilateral pair, leaving the coordination structure intact. This is the architectural advantage of a framework built on dyadic primitives: its extension to new settings does not require new framework design, only local instantiation of existing components.

7.2 Scalability and Future Directions

The bilateral structure of the Observer–Actor framework has implications for scalability that distinguish it categorically from centralized approaches. In CTDE methods, the centralized critic aggregates state and reward information from all agents simultaneously, creating a bottleneck whose computational and communication cost grows with the number of agents and whose credit assignment problem grows harder as individual contributions to joint reward become harder to isolate. In the Observer–Actor framework, each agent's computational and communication burden is bounded by the size of its local interaction neighborhood. As the population of agents grows, individual agents are not required to process additional information; the coordination structure scales by adding new bilateral interactions rather than by expanding any centralized component.

This means the framework's performance advantage over CTDE should increase with population size. When the population is small, the centralized critic's access to global information provides a learning signal that bilateral pairs cannot fully replicate. As the population grows, the cost of centralization rises while its marginal value—accurate credit assignment for individual agents in a large joint action space—falls. The crossover point at which bilateral coordination outperforms centralized coordination, and the rate at which per-agent performance in the Observer–Actor framework degrades as neighborhood size is held fixed while the population grows, are natural targets for future theoretical and empirical investigation. Establishing this crossover empirically—by fixing the task, varying population size, and measuring where CTDE degrades and where the Observer–Actor framework remains stable—would provide the clearest possible demonstration of the framework's practical significance.